MAMMAI · PARALLEL DUAL-STREAM AI DIAGNOSTIC REPORT

RPT-2503-001

Generated: 22 March 2025, 18:30 IST

PRIMARY DIAGNOSIS

⚠ Heavily Malignant

PATIENT NAME	PATIENT ID	AGE	SCAN DATE
Ananya Sharma	PT-001	45 years	20 March 2025

MODALITY	ATTENDING RADIOLOGIST
Digital Mammography (CC + MLO)	Dr. Radiologist (DR-2025-001)

🔬 Classification performed by **your locally trained PyTorch Dual-Stream model (Swin-T v2 + RAD-DINO)**. Model weights remain private. Clinical narrative generated by **Gemini 1.5 Flash LLM** based on numerical output only.

AI CONFIDENCE SCORES — 3-CLASS SOFTMAX OUTPUT

0 — Benign	1 — Lightly Malignant	2 — Heavily Malignant
12.4%	23.8%	63.8%

AI VISUAL ANALYSIS

📷 ORIGINAL MAMMOGRAM INPUT

☠️

Preprocessed 224×224 Tensor

🔥 GRAD-CAM HEATMAP (EXPLAINABLE AI)

🔴 High Activation Region

AI-GENERATED CLINICAL SUMMARY (GEMINI 1.5 FLASH)

FINDINGS

The parallel dual-stream AI model (Swin-T v2 + RAD-DINO) analyzed the submitted digital mammogram of the right breast in cranio-caudal (CC) and mediolateral oblique (MLO) projections. The Grad-CAM saliency maps reveal focal high-activation regions in the upper-outer quadrant consistent with architectural distortion and asymmetric hyperdense tissue patterns. Microcalcification clusters were noted as secondary findings.

IMPRESSION

The AI ensemble model returns a **Heavily Malignant** classification with a confidence of **63.8%**. Both encoder streams (Swin-T and RAD-DINO) independently converge on this classification, which significantly strengthens the clinical weight of this finding.

RECOMMENDATIONS

- Urgent radiologist review and confirmation is required.
- Proceed with image-guided core needle biopsy to confirm histological malignancy.
- Schedule contrast-enhanced MRI of bilateral breasts for full staging.
- Multidisciplinary oncology team referral is strongly advised.
- Notify patient and schedule follow-up within 72 hours.

⚠️ **AI-Assisted Report — Requires Radiologist Validation:** This report was generated by an artificial intelligence system and must be reviewed and countersigned by a licensed radiologist before clinical use. The AI output is intended to assist, not replace, the clinical judgement of the attending physician.

AUTHORIZATION

Dr. Radiologist
Attending Radiologist · DR-2025-001

MammAI Dual-Stream System
AI Model: Swin-T v2 + RAD-DINO · Report: Gemini 1.5 Flash